

Computer Networks



Overview of the Internet Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
 - World Wide Web and HTTP
 - FTP
 - Electronic Mail
 - DNS
- Peer-to-peer paradigm
 - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

① Introduction

- Providing Services
- Application-Layer Paradigms

② Standard Client-Server Applications

- World Wide Web and HTTP
- FTP
- Electronic Mail
- Domain Name System (DNS)

③ Peer-to-Peer Paradigm

- P2P Networks

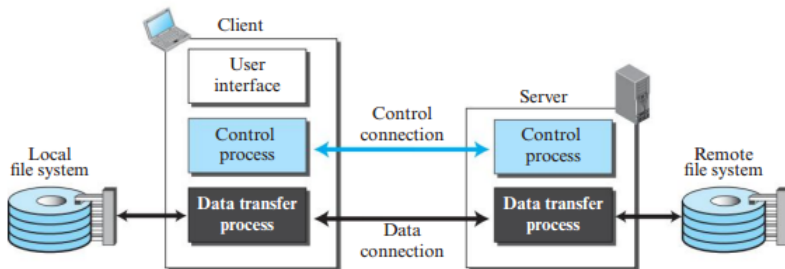
File Transfer Protocol (FTP)

- FTP stands for:
 - File Transfer Protocol
- Standard TCP/IP protocol used for:
 - Copying files between hosts
- FTP solves problems such as:
 - Different file name conventions
 - Different data representations
 - Different directory structures
- FTP provides:
 - Simple and efficient file transfer
- HTTP can transfer files, but FTP is better for:
 - Large file transfer
 - Different file formats

FTP Architecture

- FTP client contains:
 - User interface
 - Client control process
 - Client data transfer process
- FTP server contains:
 - Server control process
 - Server data transfer process
- FTP uses two separate connections:
 - Control connection
 - Data connection
- Control connection:
 - Between control processes
- Data connection:
 - Between data transfer processes

Figure 2.17 FTP



FTP Control and Data Connections

- FTP separates:
 - Commands
 - Data transfer
- Separation improves:
 - Efficiency
- Control connection characteristics:
 - Uses simple communication rules
 - Transfers command and response lines
- Data connection characteristics:
 - Uses more complex communication rules
 - Handles different data types

Lifetime of FTP Connections

- Control connection:
 - Remains open during entire FTP session
- Data connection:
 - Opened for each file transfer
 - Closed after transfer completes
- Multiple files transferred:
 - Data connection opens and closes multiple times
- FTP well-known ports:
 - Port 21 → Control connection
 - Port 20 → Data connection

FTP Control Connection

- FTP control communication uses:
 - Same approach as TELNET
- Character set used:
 - NVT ASCII
- Communication performed using:
 - Commands
 - Responses
- Suitable for control connection because:
 - One command or response sent at a time
- Each command line terminated by:
 - Carriage Return (CR)
 - Line Feed (LF)

FTP Commands

- Commands sent from:
 - FTP client control process
- Commands sent to:
 - FTP server
- Commands written in:
 - ASCII uppercase letters
- Commands may:
 - Include arguments
 - Exclude arguments
- Commands used for:
 - File operations
 - Session control
 - User authentication

Some FTP commands

Table 2.4 Some FTP commands

<i>Command</i>	<i>Argument(s)</i>	<i>Description</i>
ABOR		Abort the previous command
CDUP		Change to parent directory
CWD	Directory name	Change to another directory
DELE	File name	Delete a file
LIST	Directory name	List subdirectories or files
MKD	Directory name	Create a new directory
PASS	User password	Password
PASV		Server chooses a port
PORT	port identifier	Client chooses a port
PWD		Display name of current directory
QUIT		Log out of the system
RETR	File name(s)	Retrieve files; files are transferred from server to client
RMD	Directory name	Delete a directory
RNFR	File name (old)	Identify a file to be renamed

<i>Command</i>	<i>Argument(s)</i>	<i>Description</i>
RNTO	File name (new)	Rename the file
STOR	File name(s)	Store files; file(s) are transferred from client to server
STRU	F, R, or P	Define data organization (F : file, R : record, or P : page)
TYPE	A, E, I	Default file type (A : ASCII, E : EBCDIC, I : image)
USER	User ID	User information
MODE	S, B, or C	Define transmission mode (S : stream, B : block, or C : compressed)

FTP Responses

- Every FTP command generates:
 - At least one response
- Response contains:
 - Three-digit numeric code
 - Text explanation
- Numeric part:
 - Defines response code
- Text part:
 - Gives explanation or parameters

Some FTP Responses

Table 2.5 *Some responses in FTP*

<i>Code</i>	<i>Description</i>	<i>Code</i>	<i>Description</i>
125	Data connection open	250	Request file action OK
150	File status OK	331	User name OK; password is needed
200	Command OK	425	Cannot open data connection
220	Service ready	450	File action not taken; file not available
221	Service closing	452	Action aborted; insufficient storage
225	Data connection open	500	Syntax error; unrecognized command
226	Closing data connection	501	Syntax error in parameters or arguments
230	User login OK	530	User not logged in

Meaning of FTP Response Digits

- First digit:
 - Defines status of command
- Second digit:
 - Defines area related to status
- Third digit:
 - Provides additional information
- Response codes help client:
 - Understand server actions

FTP Data Connection

- FTP data connection uses:
 - Well-known port 20 at server
- Data connection creation differs from:
 - Control connection
- Client issues:
 - Passive open using ephemeral port
- Client sends port number to server using:
 - PORT command
- Server issues:
 - Active open using port 20

Communication over Data Connection

- Data connection used for:
 - File transfer
- Before transfer, client defines:
 - File type
 - Data structure
 - Transmission mode
- Information exchanged through:
 - Control connection
- FTP resolves heterogeneity using:
 - Common communication attributes

FTP Data Structure

- FTP supports three data structures:
 - File structure
 - Record structure
 - Page structure
- File structure:
 - Default mode
 - Continuous stream of bytes
- Record structure:
 - File divided into records
 - Used with text files
- Page structure:
 - File divided into pages
 - Each page has page number and header

FTP File Types

- FTP supports different file types:
 - ASCII file
 - EBCDIC file
 - Image file
- File type selection helps:
 - Transfer files between heterogeneous systems



FTP Transmission Modes

- FTP supports three transmission modes:
 - Stream mode
 - Block mode
 - Compressed mode
- Stream mode:
 - Default mode
 - Continuous stream of bytes
- Block mode:
 - Data transferred in blocks
 - Each block preceded by 3-byte header
- Block header contains:
 - Block descriptor
 - Block size

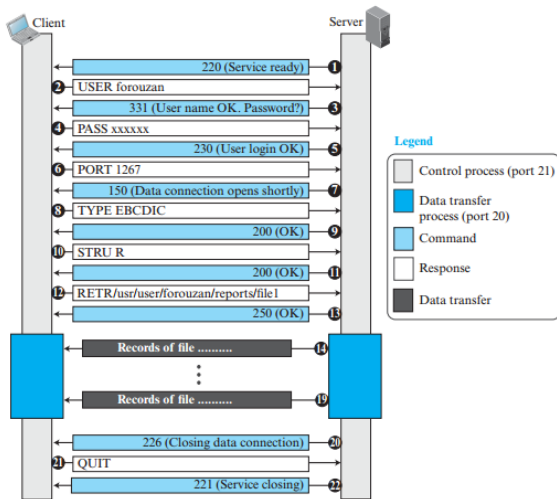
FTP File Transfer Operations

- File transfer controlled using:
 - Commands over control connection
- FTP supports three operations:
 - Retrieving file
 - Storing file
 - Directory listing
- Retrieving file:
 - Server → Client
- Storing file:
 - Client → Server
- Directory listing:
 - Server → Client



Example 2.11

Figure 2.18 Example 2.11



Example 2.11: FTP File Retrieval

- Example demonstrates:
 - Retrieving a file using FTP
- Only one file transferred:
 - From server to client
- FTP uses:
 - Control connection
 - Data connection
- Control connection:
 - Remains open during entire session
- Data connection:
 - Opened for file transfer
 - Closed after transfer

File Transfer Process in Example 2.11

- File transferred in:
 - Six sections
- Data transferred through:
 - FTP data connection
- After all records transferred:
 - Server control process sends completion message
- Client checks:
 - Whether more files are needed
- Client issues:
 - QUIT command
- QUIT command causes:
 - Control connection to close

Key Points from Example 2.11

- Control connection:
 - Persistent during FTP session
- Data connection:
 - Temporary
 - Opened and closed repeatedly
- FTP separates:
 - Commands
 - Data transfer
- QUIT command:
 - Terminates FTP session

Example 2.12

```
$ ftp voyager.deanza.fhda.edu
Connected to voyager.deanza.fhda.edu.
220 (vsFTPd 1.2.1)
530 Please login with USER and PASS.
Name (voyager.deanza.fhda.edu:forouzan): forouzan
331 Please specify the password.
Password:*****
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
227 Entering Passive Mode (153,18,17,11,238,169)
150 Here comes the directory listing.
```

drwxr-xr-x	2	3027	411	4096	Sep 24	2002	business
drwxr-xr-x	2	3027	411	4096	Sep 24	2002	personal
drwxr-xr-x	2	3027	411	4096	Sep 24	2002	school

```
226 Directory send OK.
ftp> quit
221 Goodbye.
```

Example 2.12: FTP Directory Listing Session

- Example shows:
 - Actual FTP session for directory listing
- FTP communication includes:
 - Client commands
 - Server responses
 - Data transfer
- Purpose of session:
 - Retrieve list of directories/files

FTP Session Components

- Black lines represent:
 - Commands sent by client
- Colored lines represent:
 - Responses from server control connection
- White lines with black background represent:
 - Data transfer over data connection
- FTP uses:
 - Control connection for commands/responses
 - Data connection for actual directory data

Key Points from Example 2.12

- FTP separates:
 - Control information
 - Data transfer
- Directory listing transferred through:
 - Data connection
- Commands and responses exchanged through:
 - Control connection
- FTP session demonstrates:
 - Coordination between client and server

Security Issues in FTP

- FTP designed during:
 - Early Internet era
- Original FTP design:
 - Did not focus on security
- FTP requires:
 - Username
 - Password
- Problem:
 - Password transmitted in plaintext
- Plaintext password can be:
 - Intercepted by attackers



FTP Data Transfer Security

- FTP data connection transfers:
 - Data in plaintext
- Plaintext transmission is:
 - Insecure
- Attackers may:
 - Read transferred files
 - Capture sensitive information
- Security needed for:
 - Authentication
 - Confidentiality



- Security improved by adding:
 - Secure Socket Layer (SSL)
- SSL placed between:
 - FTP application layer
 - TCP layer
- Secure version of FTP called:
 - SSL-FTP
- SSL-FTP provides:
 - Encrypted communication
 - Secure password transfer
 - Secure data transfer